ME 333 - MECHANICS OF MATERIALS
PRACTICE PROBLEMS 4 - UNSYMMETRIC BENDING & SHEAR CENTER
OCT 30, 2025

Practice Problems:
Example 7.2, From Prescribed Text - 7.2, 7.3, 7.5, 7.6, 8.1, 8.2, 8.3, 8.7.

(1) A beam consisting of steel and brass (with $E_s = 200\,\text{GPa}$ and $E_b = 100\,\text{GPa}$, respectively), bonded together to form a rectangular cross-section as shown in Fig. A, is subjected to a given moment $M_z = 25\,\text{Nm}$. Determine (a) the maximum stress in the brass and steel, (b) the flexural stress in brass and steel at the interface and (c) the radius of curvature ρ of the beam at the cross-section.

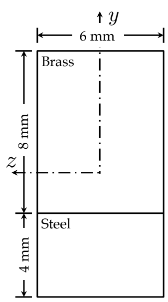

Figure A

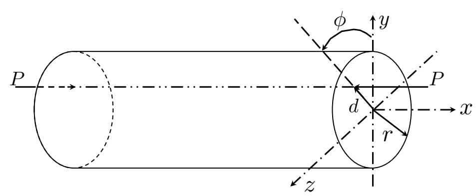

Figure B

[Answer] (a) $\sigma_{\text{brass}} = -148.3\,\text{MPa}$, $\sigma_{\text{steel}} = 211.9\,\text{MPa}$; (b) $\sigma_{\text{brass}} = 21.2\,\text{MPa}$, $\sigma_{\text{steel}} = 42.4\,\text{MPa}$; (c) $\rho = 4.72\,\text{m}$.

(2) A beam of circular cross-section with radius r is subjected to a compressive force P applied at a distance d with respect to the longitudinal x-axis as shown in Fig. B. Determine the largest value of d such that no fibers are in tension, i.e. the normal stress for all points is $\sigma_x \leq 0$.

[Answer] $d \leq \dfrac{r}{4}$

(3) The beam in Fig.1 is subjected to a force F that acts at an angle $\alpha = 30°$ to the vertical. It has a rectangular cross-section ($b \times h$, $h = 2b$). Determine the stresses at points A and B. [4 pts]

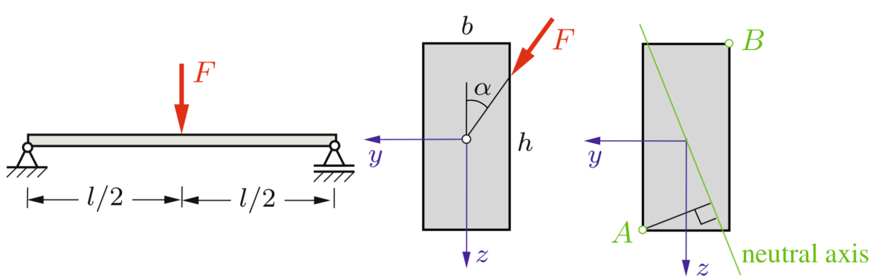

FIGURE 1. A Rectangular cross-section beam supported by ball and socket joints

[Answer]

$$\sigma_A = \frac{3Fl}{8b^3}\left(\frac{\sqrt{3}}{2}+1\right) \quad @ \quad y_A = \frac{b}{2} \text{ and } z_A = b$$

$$\sigma_B = -\frac{3Fl}{8b^3}\left(\frac{\sqrt{3}}{2}+1\right) \quad @ \quad y_A = -\frac{b}{2} \text{ and } z_A = -b$$

(4) A cantilever beam ($t << a$) carries a force F that acts in z-direction as shown in Fig.2. Determine the stresses at points C and D. [4 pts]

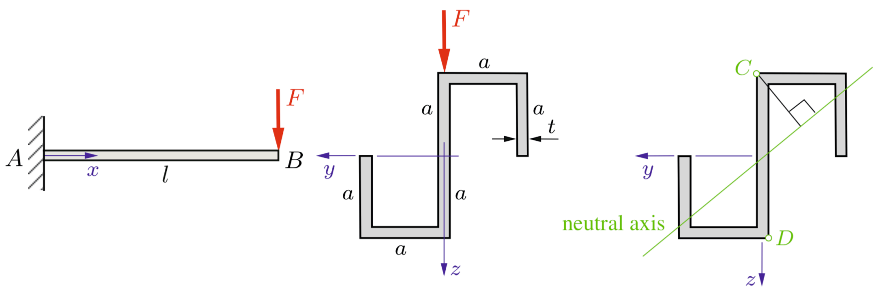

FIGURE 2. A cantilever beam with a thin-walled cross section

[Answer]

$$\sigma_C|_{\max} = \frac{6Fl}{11ta^2} \quad @ \quad y_C = 0 \text{ and } z_C = -a$$

$$\sigma_D|_{\max} = -\frac{6Fl}{11ta^2} \quad @ \quad y_D = 0 \text{ and } z_D = a$$

(5) Locate the shear center O of the thin-walled cross section ($t << r$) as shown in Fig.3. [5 pts]

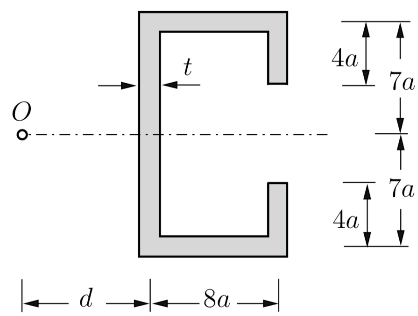

FIGURE 3. A Thin-walled cross-section

[Answer] $d = 4.85a$

Note: Do **not** use any "shortcuts" and show all the steps of your work.